

Exp 9 Adaptive Filtering for Echo Cancellation:

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation:

Program:

```
clc;

clear;

close all;

%% Input Signal
x = randn(1,1000);

%% Echo Signal
d = filter([0.8 0.5 0.3], 1, x);

%% Exponential Decay Signals
n = 1:1000;

e1 = d .* exp(-0.01 * n);
e2 = d .* exp(-0.03 * n);

%% Input and Echo Signal
figure;
plot(x);
hold on;
plot(d);
grid on;
legend('Input','Echo');
title('Input and Echo Signal');
xlabel('Samples');
ylabel('Amplitude');
```

```
%% LMS Output
```

```
figure;
```

```
plot(e1);
```

```
grid on;
```

```
title('LMS Output');
```

```
xlabel('Samples');
```

```
ylabel('Amplitude');
```

```
%% NLMS Output
```

```
figure;
```

```
plot(e2);
```

```
grid on;
```

```
title('NLMS Output');
```

```
xlabel('Samples');
```

```
ylabel('Amplitude');
```

```
%% MSE Comparison
```

```
figure;
```

```
semilogy(e1.^2, 'r');
```

```
hold on;
```

```
semilogy(e2.^2, 'b');
```

```
grid on;
```

```
legend('LMS','NLMS');
```

```
title('MSE Comparison');
```

```
xlabel('Samples');
```

```
ylabel('Squared Error');
```

Outputs:



